



Identifying knowledge boundaries

Paper proposal by Justin Cechmanek

The idea(s)

- Mainstream ideas are well represented in large, unstructured data sets
- Interesting and innovative ideas exist at the periphery of knowledge

QED, ipso facto, visa vi,

We should identify these peripheral or fringe topics as a source of new ideas

The motivation(s)

- Innovation happens at the periphery of our understanding
- Combining disparate ideas is a common ideation strategy, this formalizes it. Ex:
 - Graphics processing + statistics == GPU deep learning
 - VectorDB + NLP == agent memory
 - Reinforcement learning + recommendations == bandit algorithms
- We can identify blind spots and overlooked areas

The plan

Use modified clustering to categorize topics and identify their boundaries

Step 1

Embed a large corpus into vector space in Redis

Step 2

Standard K-means clustering to categorize data into topics

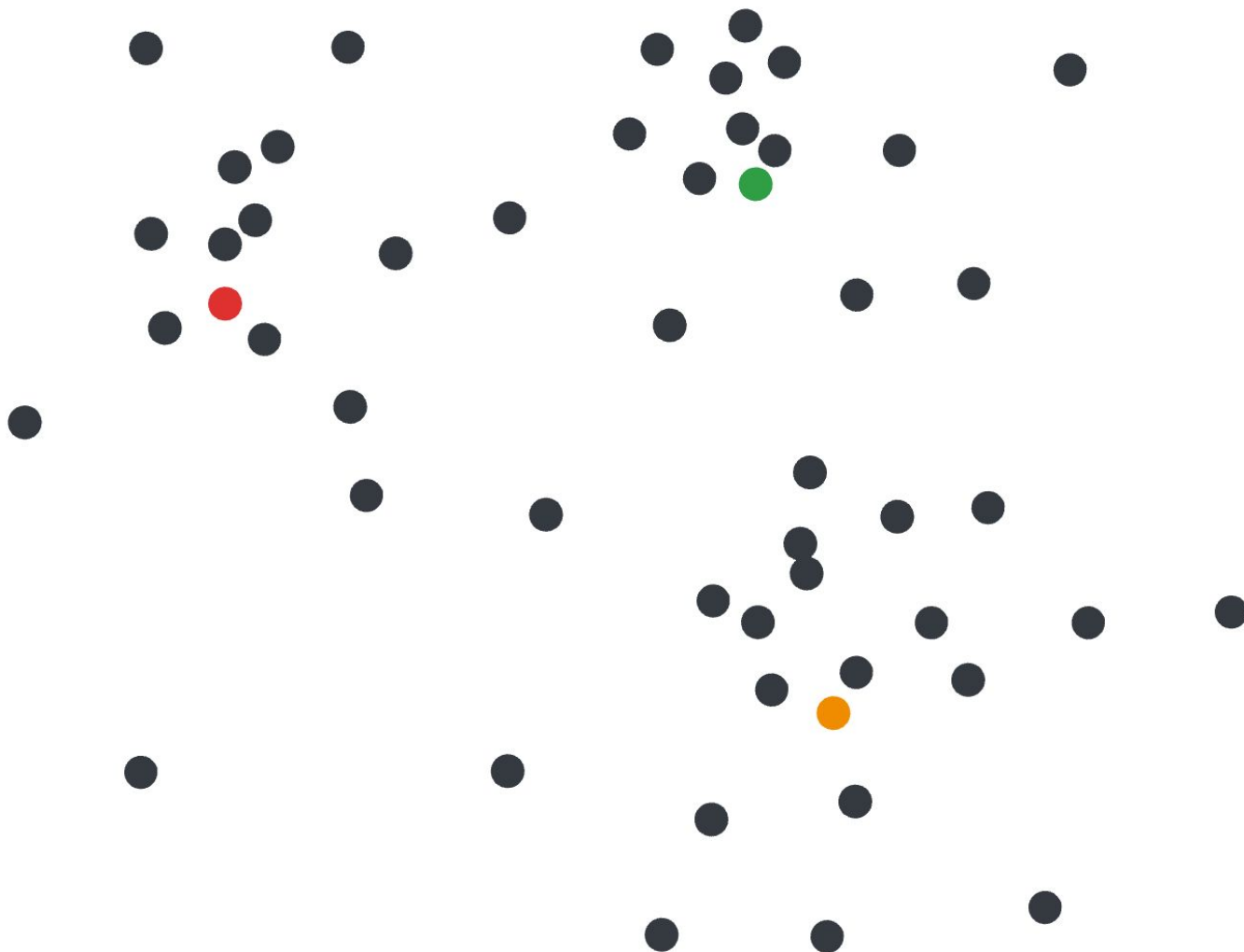
Step 3

For each cluster find the topics furthest from the centroid

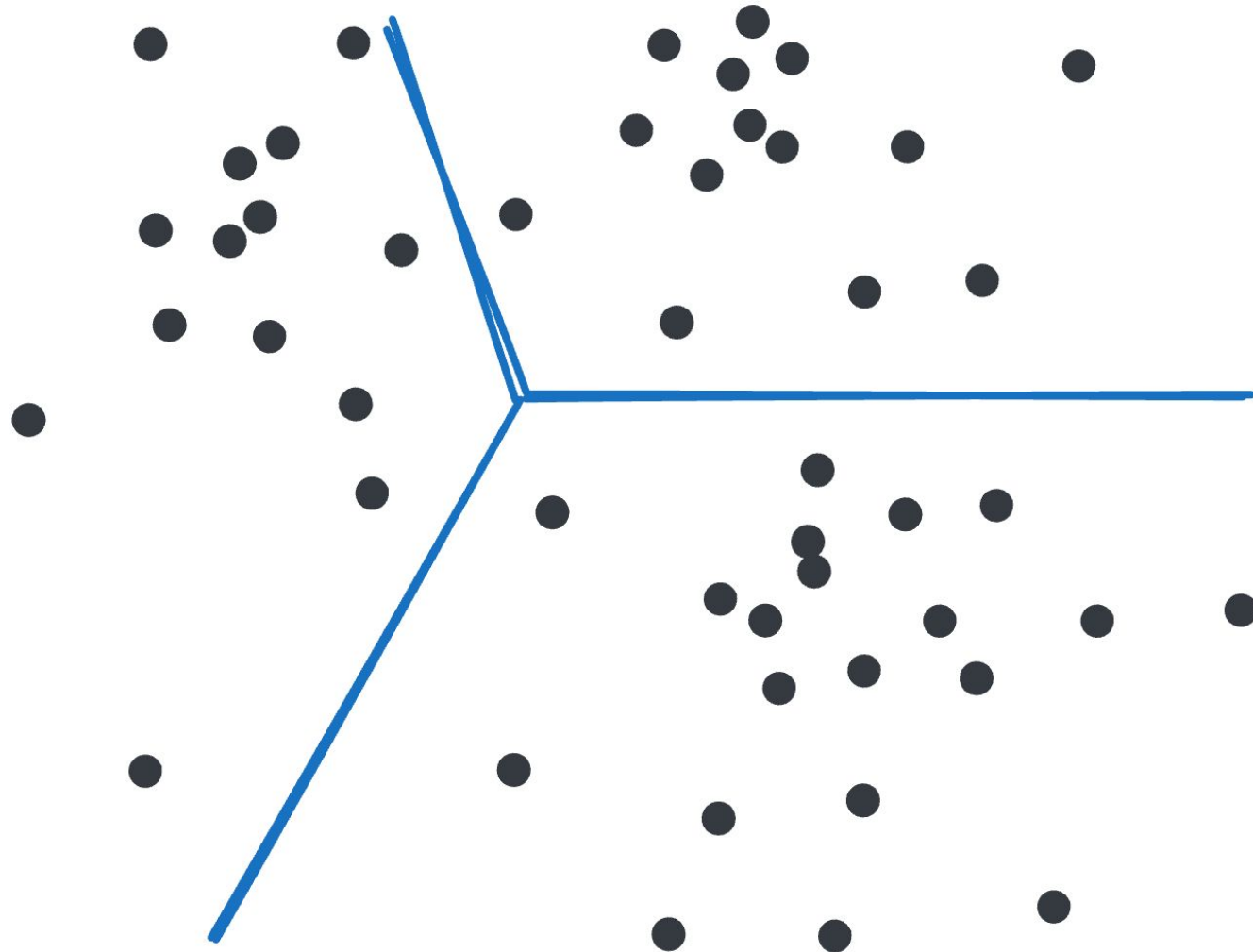
Embed data into
vector space



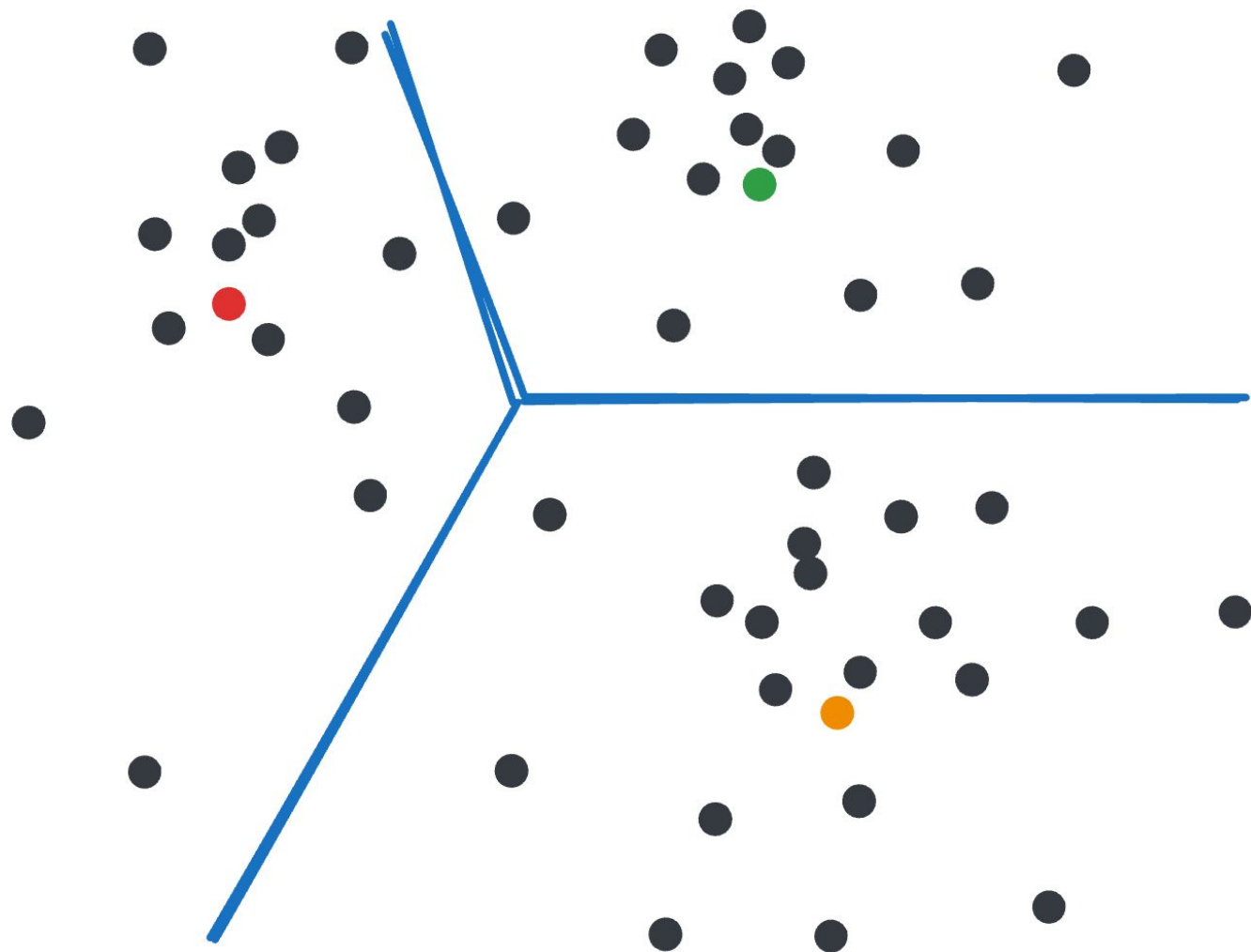
Set $K (=3)$ and
find topic
centroids

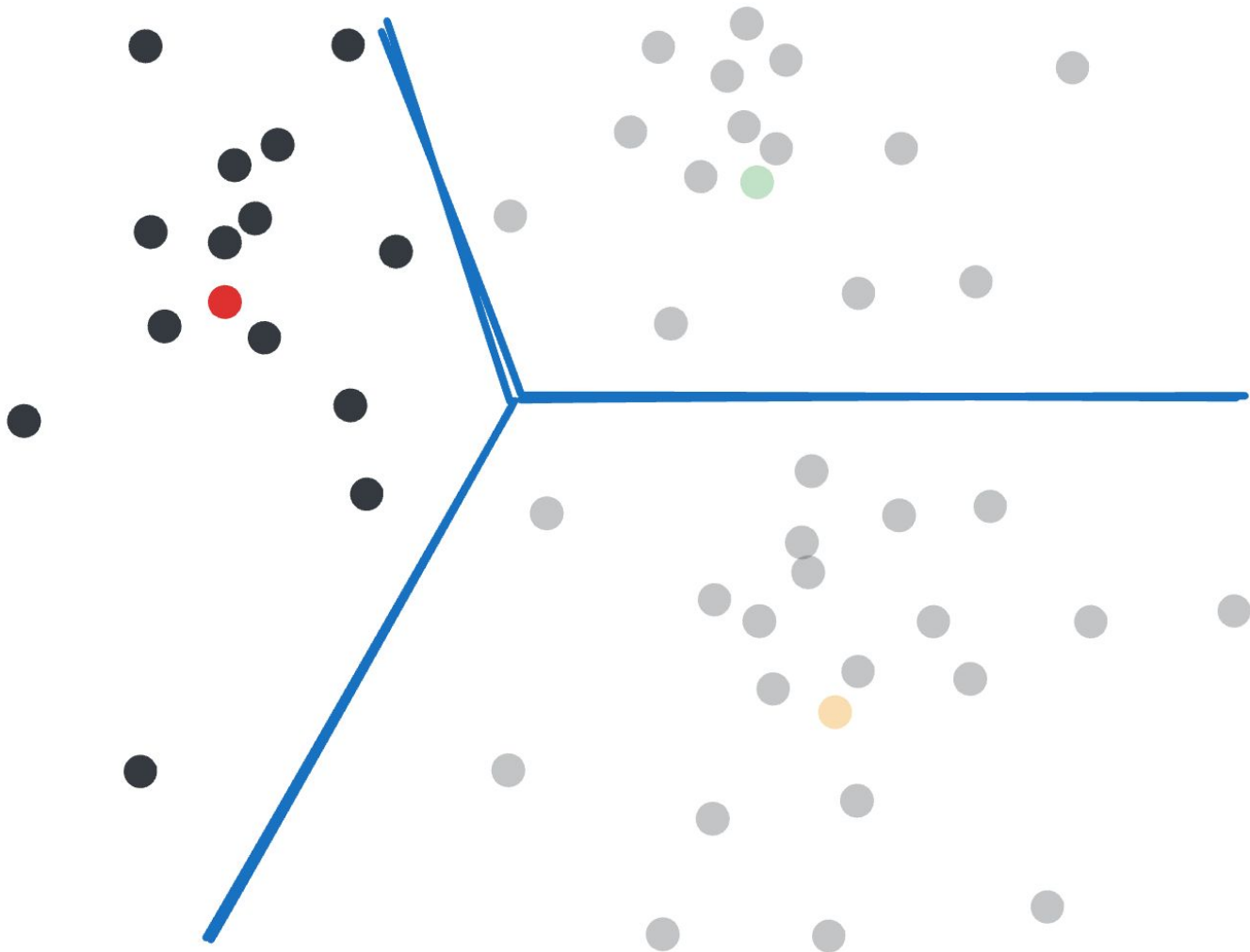


Embed data into
vector space

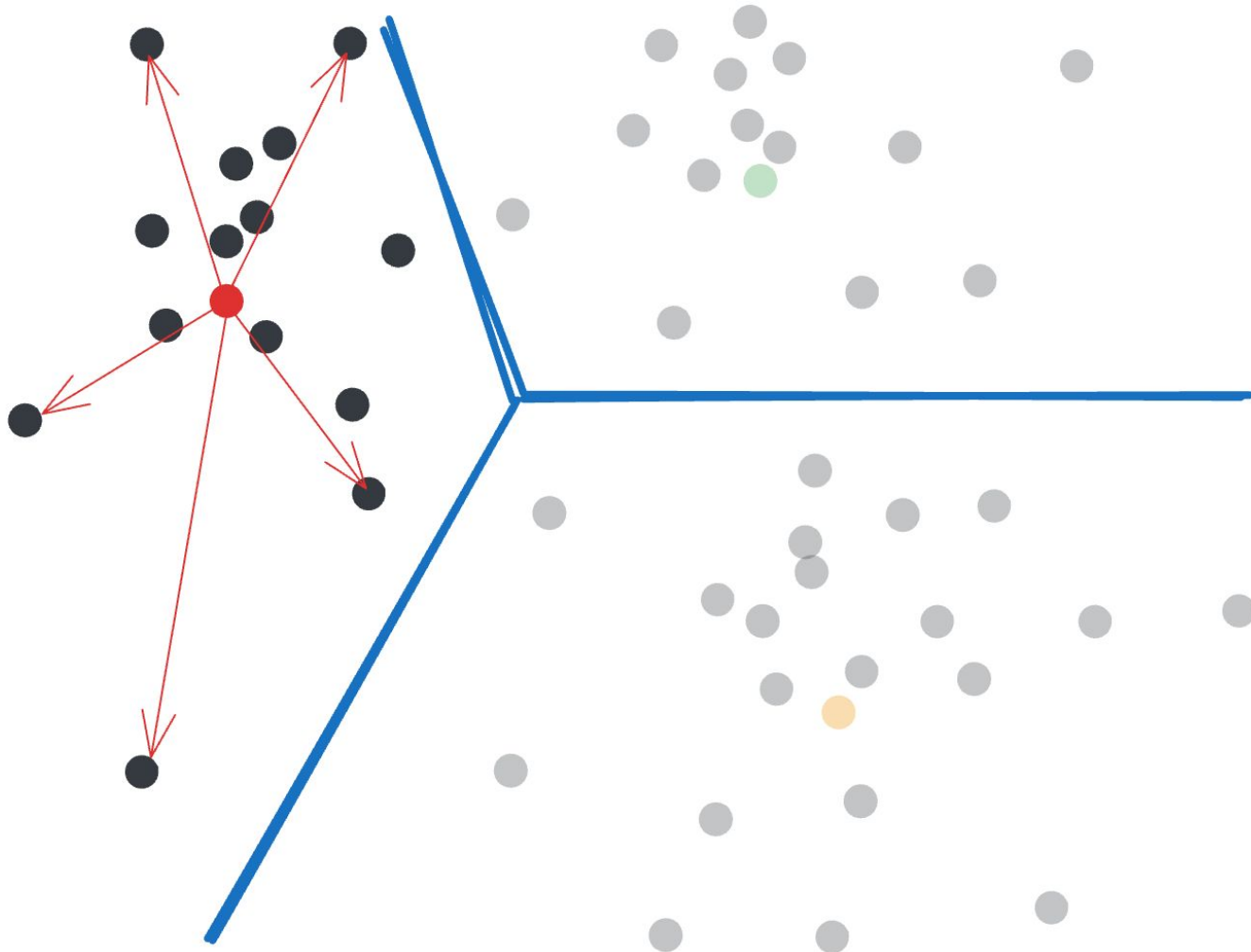


Set $K (=3)$ and
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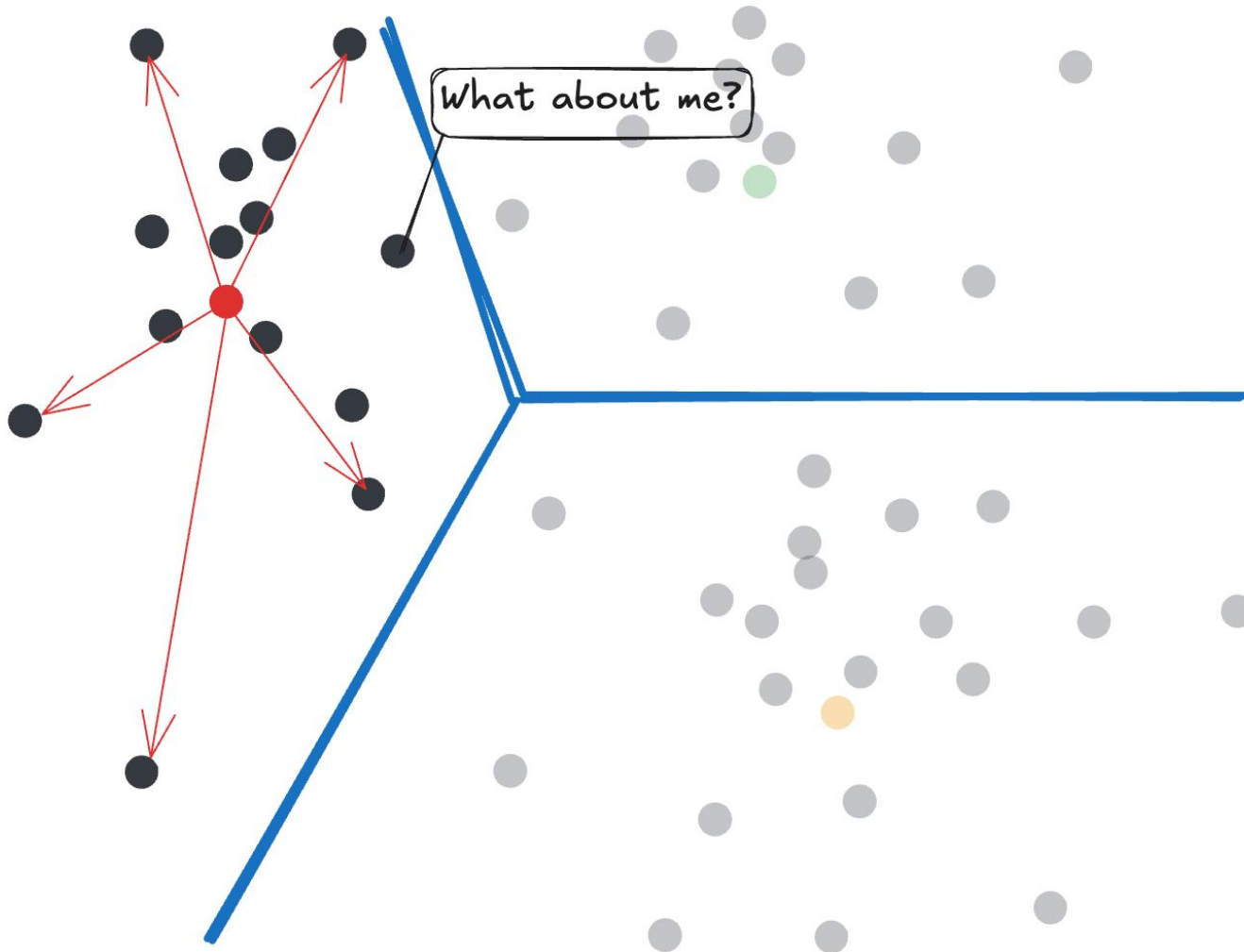




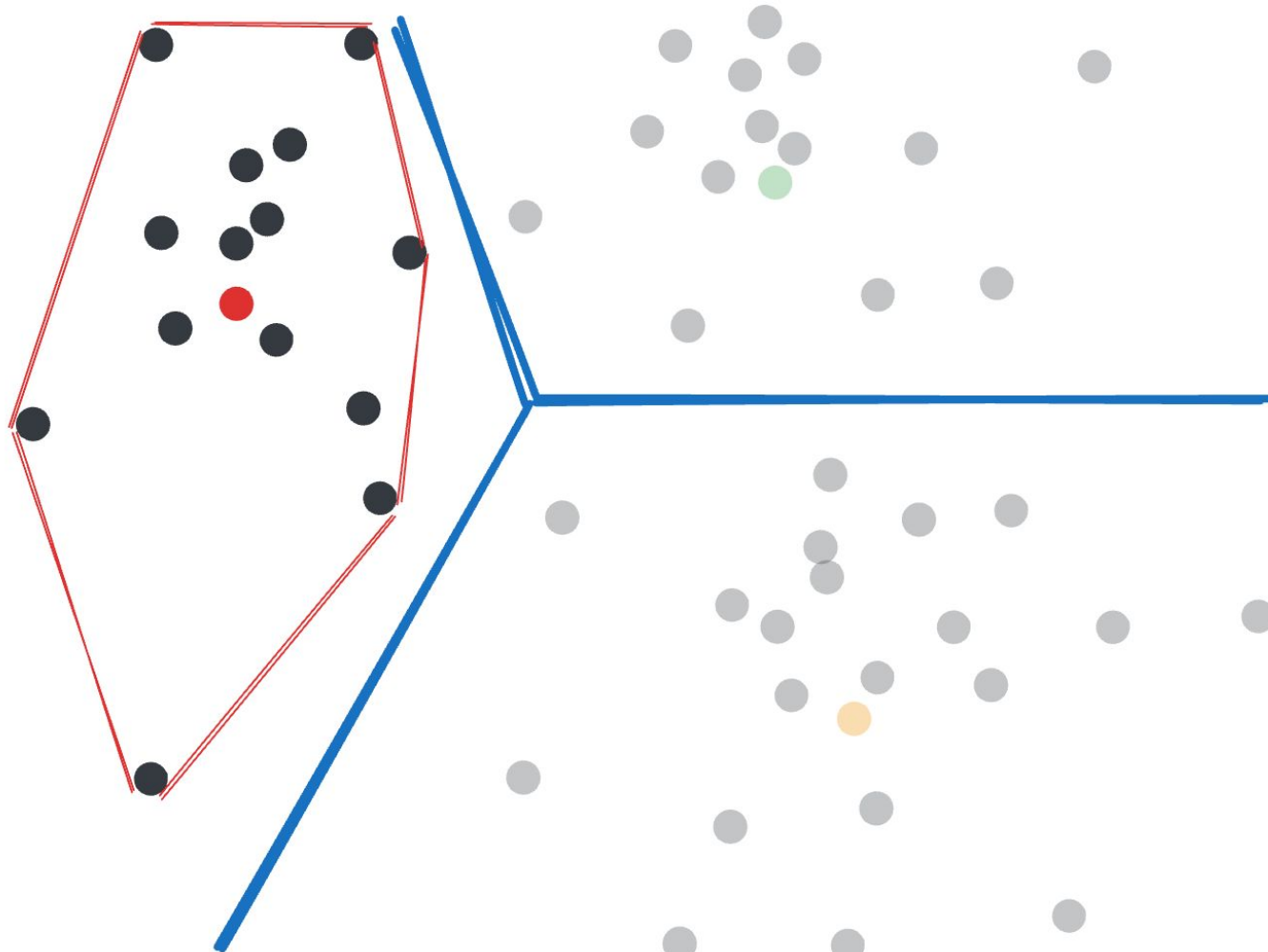
Filter down to
one cluster at a
time



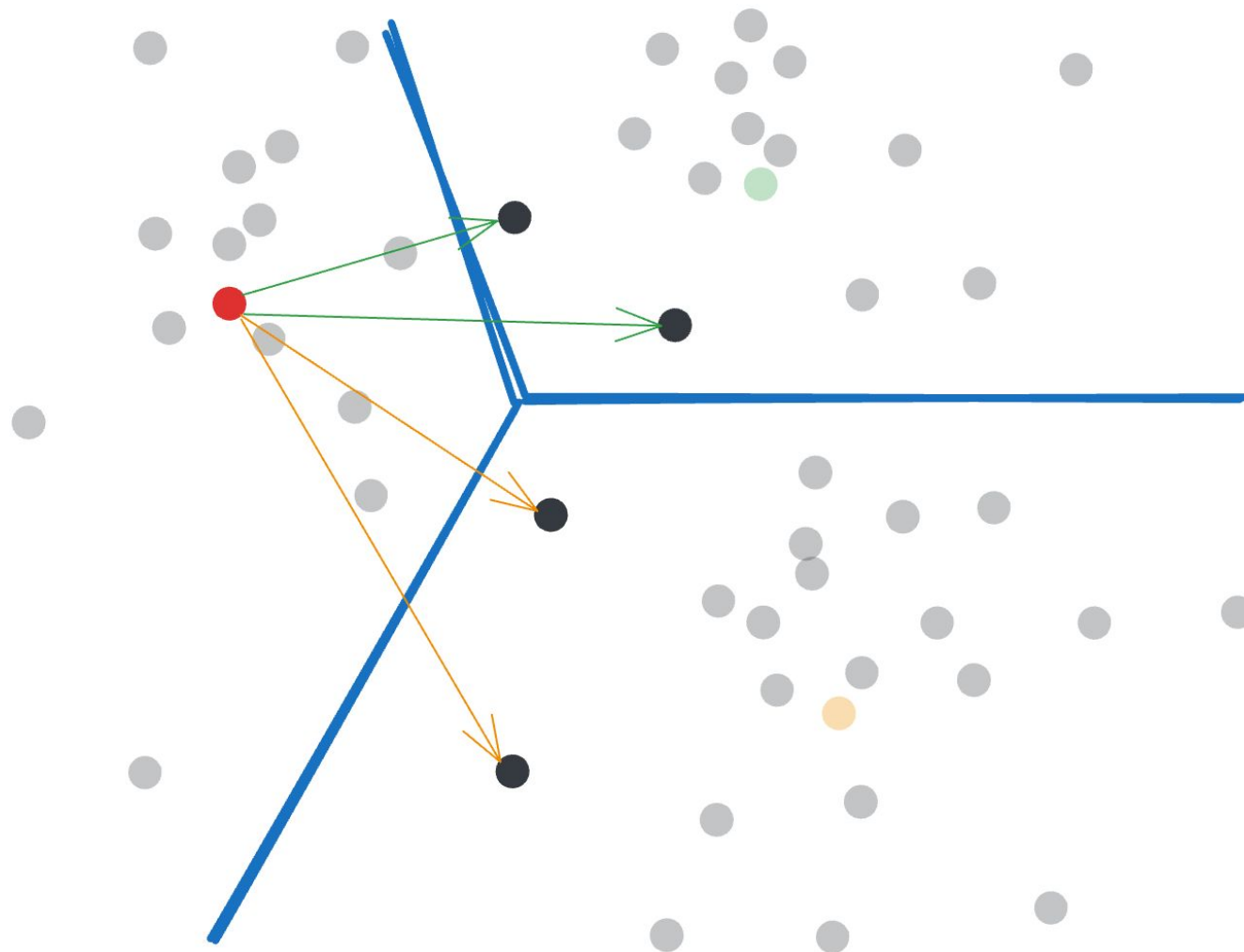
Vector range
query using
cluster centroid,
sorting by
distance
descending



Vector range
query using
cluster centroid,
sorting by
distance
descending



Find the **convex hull** of the cluster to identify topics on periphery



OR,
Find the topics
closest to a
centroid, **not** in
the centroid's
cluster.

This can
combine
disparate ideas

Tools

Redis vector search (duh)

LangChain text splitter and PyPDFLoader

Scipy spatial package, for convex hull computation

PCA for dimensionality reduction, if needed

Questions? Thoughts? Concerns? Grave Warnings?

Where may this fail?

What corpus should we start with?

How can we expand this approach to do more?

What else can we do with this?

Is this worth the time?

How do we sell this for \$\$\$\$\$\$?